

MTH1004 Probability, Statistics, and Data

Summative coursework

Submission deadline: Noon on Friday 12 December 2025

The marks for this assignment contribute 15% to the module assessment.

Instructions

Your task is to analyse and summarise data about how well people understand the meaning of words used to describe personality traits (e.g., pedantic, amiable, folksy). You will produce both numerical summaries and graphical visualisations, explain your results, and present your work in the form of a 1-page poster. This assignment assesses the Intended Learning Outcomes 3 through 7, as listed in the module descriptor.

This assignment is **AI-prohibited**. Please refer to our guidance on the use of generative artificial intelligence (GenAI) tools and to the notes on GenAI tools towards the end of this document. You may use material presented in the module (e.g., lecture notes, R lab exercises and solutions) without attribution, but any other external resources which inform your work must be properly cited, including any references mentioned in this brief (all of which are available through the university library). Follow our guidance on referencing.

This is an individual assignment and you must not discuss it with anyone other than the lecturer. Please also refer to our regulations on academic conduct and practice.

This coursework brief is structured as follows: The data is discussed in the next section and is followed by instructions about how to carry out your analysis. This is followed by a section on how to structure your submission files and the marking guidelines. More information on GenAI tools is provided in the final section. At the end of this document are Tables 1 and 2 which describe the columns in the data.

Data

You will be using two, related data sets in your analysis. Both are about the results of a study exploring how well people understand trait-descriptive adjectives (TDAs), which are words used to describe personality traits (e.g., pedantic, amiable, folksy). A list of 2,818 TDAs were assembled by the researchers and people were recruited to take 75-question multiple choice tests assessing whether they knew the definition for a subset of the TDAs on the list. This study is described further in:

Condon, D. M., Coughlin, J., and Weston, S. J. (2022). Personality trait descriptors: 2,818 trait descriptive adjectives characterized by familiarity, frequency of use, and prior use in psycholexical research. *Journal of Open Psychology Data*, 10:1, pp1-9. DOI: 10.5334/jopd.57 (link).

The file `test_results.csv` contains a subset of the results from Condon, Coughlin, and Weston (2022). Each row represents the results of one test, with columns providing information about the test taker, how long they spent taking the test, and their score. Participants were allowed to take the test multiple times and unique test-takers can be identified using the column `PID`. See Table 1 at the end of this document for more information about each column. Missing data has been encoded as `NA` in the file.

The second file, `tda_key.csv`, has 2,818 rows, one for each of the TDAs investigated in the study. Information about the proportion of times the word was correctly defined across all tests taken along with a measure of how frequently the TDA is used in English based on usage in the Google Books database is also provided in this file. See Table 2 at the end of this document for more information about each column. Missing data has been encoded as `NA` in the file.

Assignment

In the following three parts of this assignment, you will analyse data contained in `test_results.csv` and `tda_key.csv`. You can download these files from the “Summative Assessments” tile on ELE and read it into R using `read_delim()` from the package `tidyverse`. Your analysis can (and should) be completed using only those functions discussed in the lecture notes and R labs.

Part 1: Introduction

1. In your own words, describe the purpose of the study, how the study was conducted, including how participants were selected. Provide reference(s) for your answer. (No coding.)
2. For your analysis, choose only those observations from `test_results.csv` where the test takers is at least 19 years old and took at least 3 minutes to complete the TDA test. **Use this subset for the remainder of Parts 1 and 2.** How many TDA tests have been taken in this subset? Further, as participants were allowed to take the TDA test multiple times, calculate how many unique participants will be included in your analysis based on this subset.

Part 2: Comparing Participant Performance

1. Create a suitable graph to compare the participant performance on the TDA test for education levels `College/university degree (4 yr)` and `Less than 12 years`. Save your graph using `ggsave()`. Comment on the relationship between these variables including a discussion of outliers, if applicable.
2. Describe the variable `duration` using suitable summary statistics.

Part 3: Comparing Adjectives

1. Using the column `prop_item_correct` in `tda_key.csv`, identify the adjectives, if any, everyone in the study defined correctly.
2. Interpret the values in the column `gbooks_freq` in `tda_key.csv`. Explain what a positive value indicates (coding not required). Calculate the *percentage* of adjectives which are missing this variable (requires coding).
3. Create a suitable graph showing the relationship between `prop_item_correct` and `gbooks_freq` and interpret it. (If applicable, drop any relevant missing data.)

ELE Submission

Submit two files zipped together: a one-page, A4-sized pdf of your poster and your .R script through the “Report 1” ELE submission point. Do *not* include your name in the file or file names.

- **Poster Format:** Your answers to the above questions should be saved and submitted as a **one-page, A4-sized poster saved as a pdf file**. The poster can be in either landscape or portrait orientation. It must be possible to highlight and copy text on the poster when viewed in a pdf viewer (like Adobe Acrobat); that is, do **not** submit a pdf that just contains an image file of your poster. (Note: your graphs can be added to the poster as image files.) Do not submit your poster in any other file format than pdf (such as ppt, doc, png, etc.).

Your poster should be structured so as to contain an informative title and section headings corresponding to each part of the analysis. Furthermore, each section should include a brief written summary of the results and answers to all questions and graph(s) of the corresponding part. Answers that do not appear on the poster will not be marked. Write in a clear and objective tone, avoid slang or conversational language, and be understandable to someone who has not worked with the data set but knows basic statistics. You may include further information about the data or research that you think could make the poster interesting; cite any references you have used. The text on the poster should be mostly written in full sentences. As a soft guideline, aim to write between 300 and 500 words in total.

You can find more information on good formatting and writing in the “poster-hints.pdf” file posted along with this brief in the Summative Assessments tile on ELE.

- **R Code Format:** Submit all R code you used to produce your answers and figures as a **single .R script file**. It should be possible to open and run the file in RStudio without errors. The R code should be formatted for good readability, using appropriate spaces, indentation, line breaks, and code comments. You can complete this coursework using the code used throughout the lectures and lab exercises. If you use a function that was not covered in the course, add a comment on how or where you found it.
- **How to Zip Files:** Since your submitted coursework is comprised of two files, you

must zip them together to create a single, compressed file to upload to ELE. Here is how to create a zipped file: (a) [link for Windows \(e.g., a PC\)](#) or (b) [Mac](#).

Marking Criteria

This summative coursework will be worth 15% of your final module mark. Your poster and code will be marked against the following criteria, with some indicative questions that will be considered by the marker:

- **Accuracy, 40 marks:** Is the data source, along with any other references cited? (Including the author(s), publication year, and a hyperlink to the resource on the internet is a sufficient.) Are all the reported answers correct? Do the visualisations display the correct data? Do the written descriptions match the visualisations? Does the summary provide a coherent description of the analyses and results? Do the conclusions logically follow from the data analysis results? Are the reported answers relevant to the questions?
- **Coding, 30 marks:** Does the submitted R script run without errors and produce all results reported on the poster? Is the code include suitable comments and formatted for good readability? Does the code make use of functions taught in the module?
- **Writing Style, Visualisations, and Poster Design, 30 marks:** Is the written description of data and figures clear and concise? Does the text contain typos and/or grammatical errors? Are variables referred to with meaningful language as opposed to the column name/abbreviations in the data set? Are the graphs appropriately labeled (e.g., titles, units of measurement) and is the text on the graph legible? Are suitable units of measurement provided with all calculated summary statistics? Are the visualisations labelled with figure numbers and referred to in the text? Does the poster look visually attractive? Were font types, sizes, and colors sensibly chosen? Is there a suitable title and section headings?
- **Creativity Bonus, up to +10 marks possible:** A bonus of up to 10 marks can be awarded if the report contains additional interesting data analyses that were not explicitly asked in any of the questions and/or if the visualisations are particularly well-done. (Note: The total coursework mark cannot exceed 100 marks.)

Statement on the Use of GenAI Tools

The University of Exeter is committed to the ethical and responsible use of Generative AI (GenAI) tools in teaching and learning, in line with our academic integrity policies. Direct copying of AI-generated content can be an academic offence and is addressed under plagiarism, misrepresentation and contract cheating in Chapter 12.3 of the Teaching Quality Assurance Manual.

This assessment is **AI-prohibited**. This means that you must not use AI tools under any circumstances when preparing and writing for this assessment. This is because using

AI tools prevents fair assessment of your ability to demonstrate the module learning outcomes. When submitting your assessment, you must do the following:

Check the box during the submission process that confirms you have adhered to the university's academic conduct policy and the expectations on use of GenAI in your assessment brief.

Table 1: Description of columns in `test_results.csv`; missing data has been encoded as NA in the data file.

	name	type	description
1	PID	text	unique test-taker (i.e., participant) identifier
2	duration	numeric	length of time in seconds taken to complete TDA test
3	age	text	self-identified age group (categories: 18andUnder, 19to24, 25to29, 30to34, 35to39, 40to49, 50to59, 60andOver)
4	sex	text	self-identified sex (categories: Female, Male, Other)
5	edu	text	self-identified education level (categories in order: Less than 12 years, Associate degree (2 yr), Currently in college/university, College/university degree (4 yr), Currently in grad/professional school, Grad/professional degree)
6	prop_correct	numeric	proportion of correct answers in TDA test

Table 2: Description of columns in `tda_key.csv`; missing data has been encoded as NA in the data file.

	name	type	description
1	adjective	text	trait descriptive adjective (TDA) tested in study
2	prop_item_correct	numeric	proportion of tests in which the question about this adjective was defined correctly
3	gbooks_freq	numeric	The values in this column are calculated as follows: (i) A list of the 50,000 most frequently used words was assembled using the Google Books database; (ii) For word i on this list, the number of times word i appeared in the Google Books database was counted, call this f_i ; (iii) The mean ($\bar{x}$) and standard deviation (s) of $f_1, \dots, f_{50000}$ was calculated; and (iv) For each word on the list, f_i is converted into a z -score (i.e., $z_i = (f_i - \bar{x})/s$). Therefore, the values in this column are either the z -score for the TDA or NA. The latter indicates the TDA is not used frequently enough to appear on the initial 50,000 word list compiled in step (i).